

$$\theta_j^0 := \theta_j^0 - \alpha \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

hypothesis func. ground truth

(for every j)

$\hookrightarrow j$ work till point d

meaning ps $\theta_0, \theta_1, \theta_2, \dots, \theta_d$

$$\theta_0 = \theta_0 - \alpha \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_0^{(i)}$$

$$\theta_1 = \theta_1 - \alpha \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_1^{(i)}$$

$$\theta_2 = \theta_2 - \alpha \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_2^{(i)}$$

Numerical Example

x_0	x_1	x_2	y
1	1	2	3
1	2	1	4
1	3	3	5

$$h = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

Ques Apply GD and compute new value of θ after two iterations

$$\alpha = 0.1$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

$$n = 3 \quad \begin{bmatrix} \theta_0 & \theta_1 & \theta_2 \\ -0 & -0 & -0 \end{bmatrix}$$

$$X = \mathbb{R}^{d+1}$$

LAB - 3 Linear Regression

14/01

- may be miss global minima if we have larger α
- Advantage is that we reach fast to global minima

Solution of Numerical:

Formula:

$$h_{\theta} = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

$$\theta_j^0 := \theta_j^0 - \alpha \sum_{i=1}^n (h_{\theta}(x^{(i)}) - (y^{(i)})) x_j^{(i)}$$

Iteration 1

$$\theta_0 = \theta_1 = \theta_2 = 0$$

$$\text{updated } \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

$$h_{\theta}(x^{(i)}) = 0$$

errors:

i	$h - y$
1	$0 - 3 = -3$
2	$0 - 4 = -4$
3	$0 - 5 = -5$

$$\theta_0 := 0 - 0.1 \sum_{i=1}^3 ((\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2) (1+2+3)) - (12))$$

$$= 0 - 0.1 \sum_{i=1}^3 0 - 12$$

$$= -0.1 \times -12$$

$$\theta_0^{(1)} = 1.2$$

$$\theta_1 := \theta_1 - \alpha \sum_{i=1}^{n-1} [h_0(x^{(i)}) - y^{(i)}] (x_1^{(i)})$$

$$\theta_j^0 := \theta_j^0 - \alpha \sum_{i=1}^n (h_0(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

$$h_0 = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

$$= (0.2)(1) + 0 + 0$$

$$h_0 = 1.2$$

$$\theta_1 = 0 - 1.2 \sum_{i=1}^{n=3} [1.2 - y^{(i)}] x_1^{(i)}$$

$$\theta_1 = 0 - 0.1 \sum_{i=1}^{n=3} [0 - (12)] 2$$

$$\theta_1 = 0 - 0.1 \sum_{i=1}^{n=3} [0 - (-24)]$$

$$\theta_1 = 0 - 0.1 \times -24$$

$$\theta_1 = 2.4$$

update $\theta_1 \rightarrow \sum (h - y) x_1 = (-3)(1) + (-4)(2) + (-5)(3)$

$$= -3 - 8 - 15 = -26$$

update $\theta_2 \rightarrow \sum (h - y) x_2 = (-3)(2) + (-4)(1) + (-5)(3)$

$$= -6 - 4 - 15 = -25$$

$$\theta_2^{(1)} = 0 - 0.1(-25) = 2.5$$

$$\rightarrow \theta^{(1)} = \begin{bmatrix} 1.2 \\ 2.6 \\ 2.5 \end{bmatrix}$$

II Iteration - 2

$$h = 1.2 + 2.6x_1 + 2.5x_2$$

i	$h(x)$	y	$h - y$
1	$1.2 + 2.6(1) + 2.5(2) = 8.8$	3	5.8
2	$1.2 + 2.6(2) + 2.5(1) = 8.9$	4	4.9
3	$1.2 + 2.6(3) + 2.5(3) = 16.5$	5	11.5

$$\theta_0 = \frac{\sum (h^{(2)} - y^{(2)}) x_0^{(2)}}{\sum (h^{(2)} - y^{(2)})} = \frac{\sum (5.8 + 4.9 + 11.5) \cdot 1}{\sum 1} = \frac{22.2}{3} = 7.4$$

$$\theta_0^{(2)} = 1.2 - 0.1(22.2) = -1.02$$

$$\theta_1 = \frac{\sum (h - y) x_1}{\sum (h - y)} = \frac{(5.8)(1) + (4.9)(2) + (11.5)(3)}{5.8 + 4.9 + 11.5} = \frac{50.1}{22.2} = 2.256$$

$$\theta_1^{(2)} = 2.6 - 0.1(50.1) = -2.41$$

$$\theta_2 = \frac{\sum (h - y) x_2}{\sum (h - y)} = \frac{(5.8)(2) + (4.9)(1) + (11.5)(3)}{5.8 + 4.9 + 11.5} = \frac{51.0}{22.2} = 2.297$$

$$\theta_2^{(2)} = 2.5 - 0.1(51.0) = -2.6$$

after 2 iterations:

$$\theta = \begin{bmatrix} -1.02 \\ -2.41 \\ -2.6 \end{bmatrix}$$

$$\begin{bmatrix} 0.1 \\ 0.1 \\ 0.1 \end{bmatrix}$$

$$\theta = \begin{bmatrix} -1.02 \\ -2.41 \\ -2.6 \end{bmatrix}$$

Ques

x_1	x_2	y
1	2	5
3	4	11

$$\theta = (X^T X)^{-1} X^T Y$$

$$\theta = ?$$

Solⁿ

$$X =$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\theta =$$

$$\begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} \quad (d+1) \times 1$$

$$Y =$$

$$\begin{bmatrix} 5 \\ 11 \end{bmatrix} \quad (2 \times 1)$$

$$\nabla_x (x^T A x) = 2 A x \text{ (identity)}$$

Step - 1 $X^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $y^{(1)} = 5$

$X^{(2)} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$; $y^{(2)} = 11$

$X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (2x2) $\vec{y} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$ (2x1)

$\theta = (X^T X)^{-1} X^T \vec{y}$

Step - 2

$X^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 11 \end{bmatrix}$

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$= \begin{bmatrix} 1 \times 1 + 3 \times 3 & 1 \times 2 + 3 \times 4 \\ 2 \times 1 + 4 \times 3 & 2 \times 2 + 4 \times 4 \end{bmatrix}^{-1} \begin{bmatrix} 1 \times 5 + 3 \times 11 \\ 2 \times 5 + 4 \times 11 \end{bmatrix}$

$= \begin{bmatrix} 1+9 & 2+12 \\ 2+12 & 4+16 \end{bmatrix}^{-1} \begin{bmatrix} 5+33 \\ 10+44 \end{bmatrix}$

$= \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}^{-1} \begin{bmatrix} 38 \\ 54 \end{bmatrix}$

$= \begin{bmatrix} 20 & -14 \\ -14 & 10 \end{bmatrix} \begin{bmatrix} 38 \\ 54 \end{bmatrix} = \begin{bmatrix} 20 \times 38 + (-14) \times 54 \\ -14 \times 38 + (10) \times 54 \end{bmatrix}$

31/01

	x_1	x_2	y
1	1	0	0
2	2	1	0
3	3	1	0
4	4	2	1
5	5	3	1

$$\theta_1 = 0, \theta_2 = 0$$

$\alpha = 0.1$
no. of samples, $m = 5$

logistic hypothesis,

$$h_{\theta}(x) = g(z) = \frac{1}{1 + e^{-z}}$$

Gradient Formula:

$$\theta_j^0 := \theta_j + \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

$$h_{\theta}(x) = g(\theta^T x)$$

$$z = \theta^T x$$

Iteration: -1

$$\theta_j^0 := \theta_j + \alpha \sum_{i=1}^m (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)}$$

① Compute hypothesis

$$\theta_1 = \theta_2 = 0$$

$$z = \theta^T x \Rightarrow 0$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^0} = \frac{1}{1+1} = \frac{1}{2}$$

$$h = \frac{1}{2} \text{ or } 0.5$$

② Compute gradient

$$\theta_1 \Rightarrow \sum_{i=1}^5 (h_{\theta}(x^{(i)}) - y^{(i)}) x_1^{(1 \dots 5)}$$

here $x \rightarrow 1$ $y \rightarrow 0$ $\sum_{i=1}^5 (y^{(i)} - h_{\theta}(x^{(i)})) x_1^{(1 \dots 5)}$

i	$h - y$	x_1	$(h - y)x_1$
1	$0.5 - 0 = 0.5$	1	$0.5 \times 1 = 0.5$
2	$0.5 - 0 = 0.5$	2	$0.5 \times 2 = 1$
3	$0.5 - 0 = 0.5$	3	$0.5 \times 3 = 1.5$
4	$0.5 - 1 = -0.5$	4	$-0.5 \times 4 = -2.0$
5	$0.5 - 1 = -0.5$	5	$-0.5 \times 5 = -2.5$

$$\sum (h - y)x_1 = 0.5 + 1 + 1.5 + (-2.0) + (-2.5) = +1.5$$

$$\theta_1 = \theta_0 + \alpha(+1.5)$$

$$= 0 + 0.1 \times +1.5$$

$$\theta_1 = +0.15$$

Iteration = 2

$$Z = 0.15x_1 + 0.15x_2$$

i	$z^{(i)}$	$h^{(i)}$
1	0.15	0.5374
2	0.45	0.6106
3	0.60	0.6457
4	0.90	0.7109
5	1.20	0.7685

θ_1	i	$y - h$	x_1	$(y - h)x_1$
	1	-0.5374	1	-0.5374
	2	-0.6106	2	-1.2212
	3	-0.6457	3	-1.9371
	4	+0.2891	4	+1.1564
	5	+0.2315	5	+1.1575

Summation $\rightarrow -1.3818$

$$\theta_1 = 0.15 + 0.1(-1.3818)$$

$$= 0.0118$$

θ_2

i	$y^{(i)} - h^{(i)}$	$x_2^{(i)}$	$(y-h)x_2$
1	-0.5374	0	0
2	-0.6106	1	-0.6106
3	-0.6457	1	-0.6457
4	+0.2891	2	+0.5782
5	0.2315	3	+0.6945

Summation $\Rightarrow 0.0164$

$$\theta_2 = 0.15 + 0.1(0.0164)$$

$$= 0.1516$$

10/02 Tree Based Algorithm

- Interpretability

① Decision Tree [Binary]

① Classification Tree

Ayclic graph $\rightarrow$ Tree $\rightarrow$ Nodes and leaf leaves

- internal nodes
- leaf

Min Systolic blood pressure in 24h period ≥ 91

NO

HR

Yes

Age ≥ 65

NO

LR

Yes

Is Sinus tachycardia present

No

LR

Yes

HR

HR : High Risk

LR : Low Risk

Labels : LR, HR

- we reads labels from leaves

leaf

$$f(x) \Rightarrow \hat{y}$$

• Features :

x_1 : date

x_2 : age

x_3 : height

x_4 : weight

x_5 : Sinus tachycardia

x_6 : minimum systolic blood pressure 24 hour

x_7 : latest diastolic blood pressure

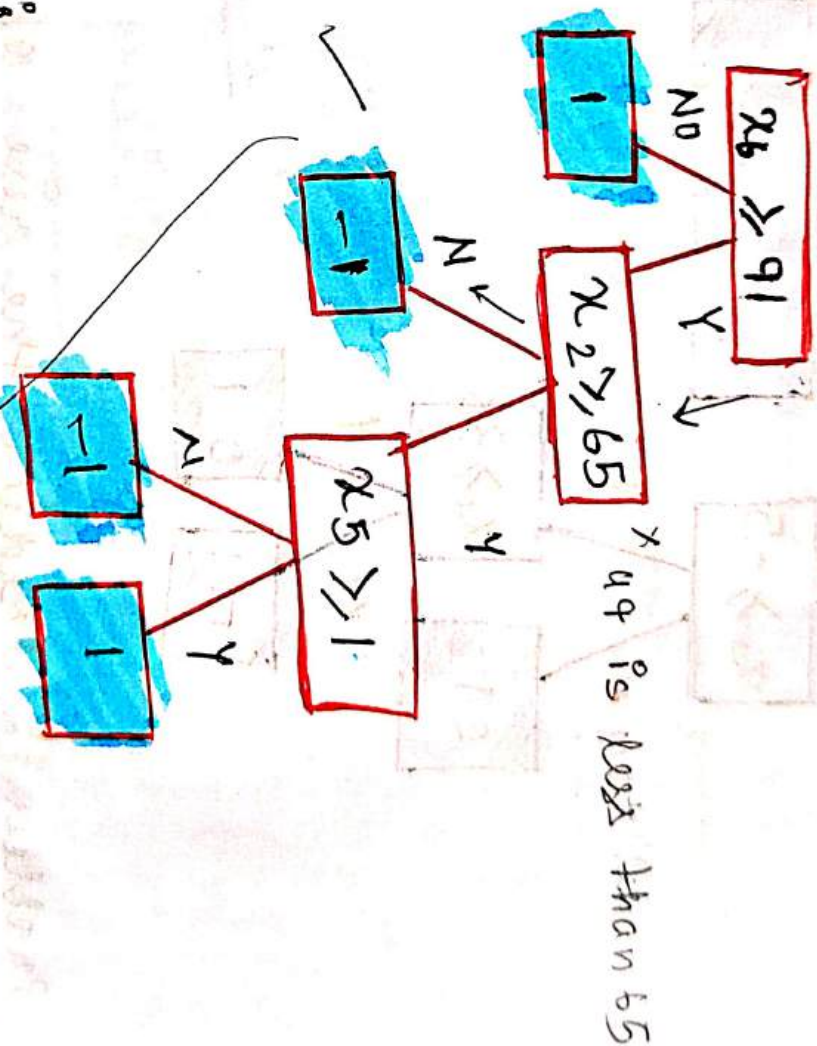
① Internal Node :

- dimension index , j
- split value , S (91)

(6)

- two child nodes
 - internal (2 component: split, dimension)
 - leaf.

Model: T



Unseen sample:

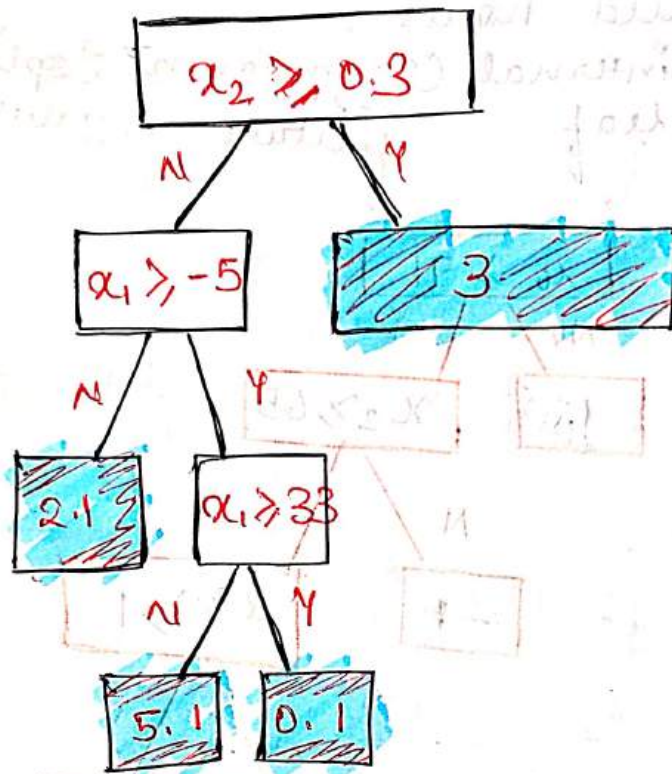
$x^{(1)}$ 2020/11/17,

49, 172 cm,

70.5 kg, 0, 115, 79

$T(x^{(1)}) = ?$

• leaf has classification
kinda value (-1, 1 in
this case) i.e. called
classification tree



11/02

Problem : Predict the rise of Autism

x_1 IQ Score	x_2 Social Score	x_3 Verbal Score	x_4 Risk
100	79	86	Low
30	40	70	High
80	85	65	Low
90	82	45	High

Test Sample:

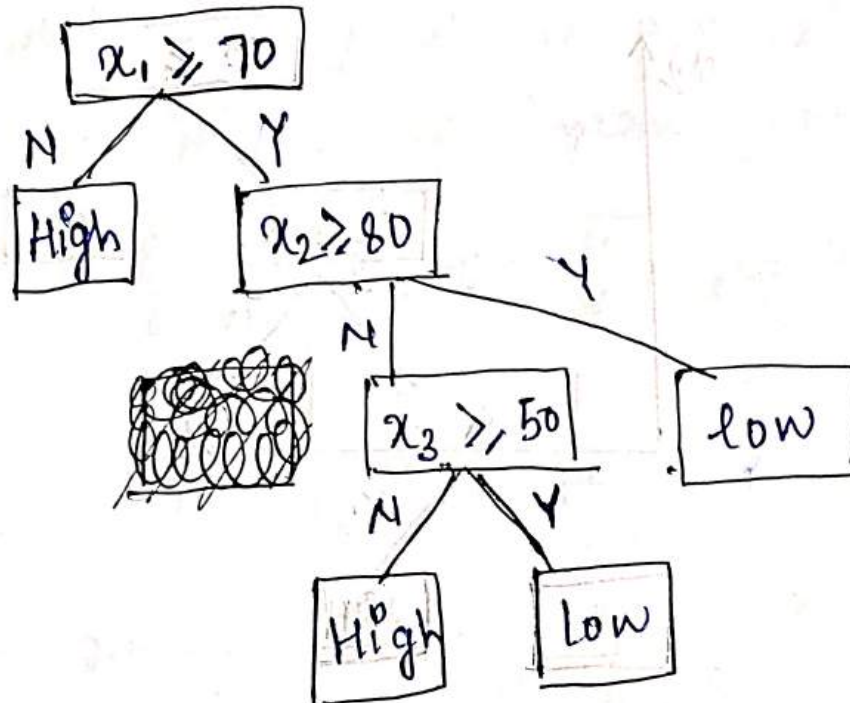
$x^{(1)}$ test $\Rightarrow$

= IQ = 120

Social = 60

Verbal = 40

Risk for
Autism } ?



(ii) Regression tree

- House price prediction

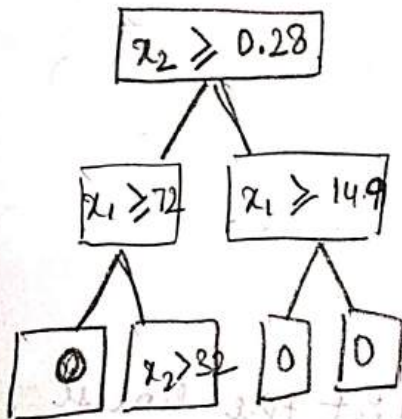
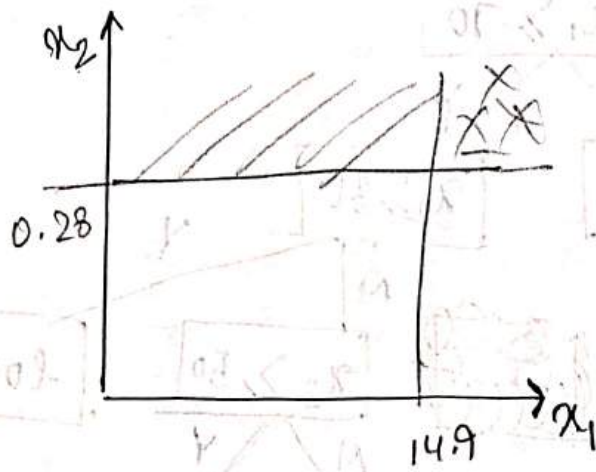
Area	Bedroom	Price
1700	4	60
1500	2	30
1200	3	40
1100	1	20

Predict the house price

- Area = 1200
- Bedroom = 2

Parameters

- split dimension
- split value
- child nodes



12/02

Sample

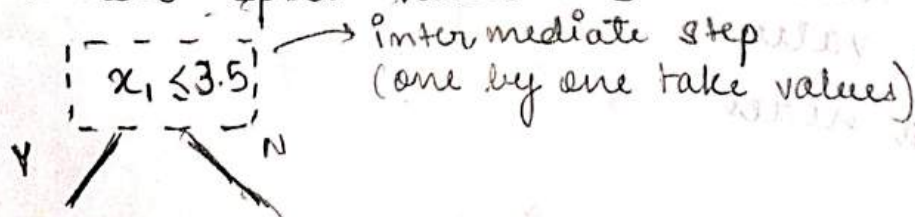
	x_1	x_2	x_3	y
1	2	3	1	5
2	1	2	2	4
3	3	1	2	6
4	4	3	3	9
5	5	2	1	10
6	6	1	2	12

(1) Take features x_1

(2) Sort feature values
[1, 2, 3, 4, 5, 6]

(3) Candidate split
points (middle value
of 2 pairs)
[1.5, 2.5, 3.5, 4.5, 5.5]

for x_1 , candidate split value 3.5



Sample 1 $x_1 = 2, 1, 3$
 Sample 2 $y = 5, 4, 6$
 Sample 3 $y = 6$

Sample 4 $x_1 = 4, 5, 6$
 Sample 5 $y = 9, 10, 12$
 Sample 6

ground truth $\rightarrow y = [5, 4, 6]$

predicted value $\rightarrow$ mean = $\frac{5+4+6}{3} = 5$

~~MSE = $\frac{(5-5)^2 + (4-5)^2 + (6-5)^2}{3}$~~

MSE = $\frac{(5-5)^2 + (4-5)^2 + (6-5)^2}{3}$

MSE = 0.67

ground truth $\rightarrow y = [9, 10, 12]$

predicted value $\rightarrow$ mean = $\frac{9+10+12}{3} = 10.33$

mean square error (MSE) = $\frac{(9-10.33)^2 + (10-10.33)^2 + (12-10.33)^2}{3}$

MSE = 1.56

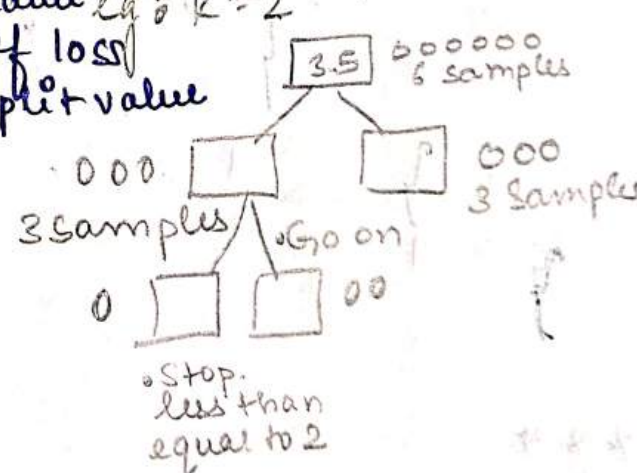
Total Error $\rightarrow 0.67 + 1.56 = 2.2$

Feature	Candidate Split Value	Loss
x_1	1.5	l_1
	2.5	l_2
	3.5	2.2
	4.5	
	5.5	
x_2	c_1	l_6
	c_2	l_7
	c_3	l_4
x_3		

$k \rightarrow$ hyperparameter

when to stop
less than this the element is leaf.

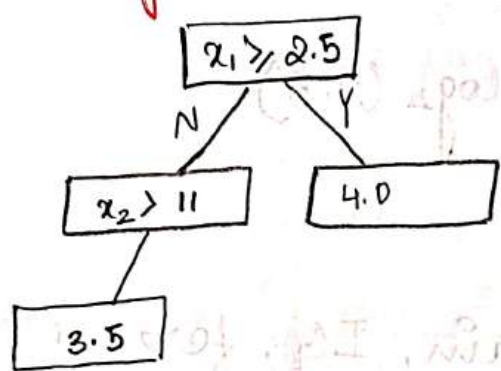
will take lower value eg $k=2$ of loss split value



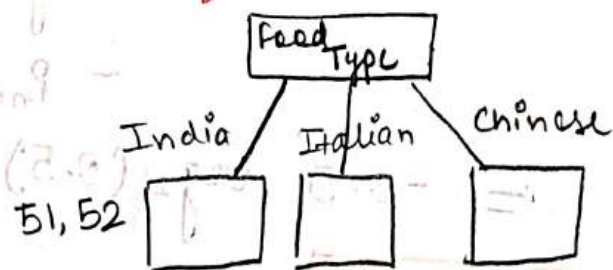
16/02

Decision Trees

Regression Tree



Classification Tree



	WEATHER	WINDY	PLAY TENNIS
S1	Sunny	No	No
S2	Sunny	Yes	No
S3	Overcast	No	Yes
S4	Rain	No	Yes
S5	Rain	Yes	No
S6	Rain	No	Yes
S7	Overcast	Yes	Yes
S8	Sunny	No	No

Heterogeneous → 0000
 XXXX
 Homogeneous → 0000

idea is to make entropy very low when we reach last leaf

Step-1. Compute root entropy

Yes = 4, No = 4

$P(Yes) = \frac{4}{8} = 0.5$

$P(No) = \frac{4}{8} = 0.5$

Entropy, H

$H_{root} = \sum -P_i \log_2 P_i$

impurity/purity
 2 class
 * can take any number of classes for computing entropy

$$\begin{aligned}
 &= -P_{yes} \log_2 P_{yes} + \\
 &\quad - P_{no} \log_2 P_{no} \\
 &= -0.5 \log_2 (0.5) + -0.5 \log_2 (0.5)
 \end{aligned}$$

$$H_{weather} = 1$$

Step 2: Compute Information Gain, IG, for
feature = weather

Weather has 3 values, Sunny, Overcast, Rain

Sunny:

3 samples, S_1, S_2, S_3 , (No, No, No) $P(\text{No}) = 3/3 = 1$

$$H_{\text{sunny}} = \sum_{i=1}^1 -P_i \log_2 P_i$$

$$= -1 \log_2 1$$

$$H_{\text{sunny}} = 0 \Rightarrow \text{Entropy is zero, zero}$$

impurities $\Rightarrow$ Purity $\rightarrow$

Homogeneous

Overcast:

2 samples S_3, S_7 (Yes, Yes) $P(\text{Yes}) = 2/2 = 1$

$$H_{\text{overcast}} = \sum_{i=1}^1 -P_i \log_2 P_i$$

$$= -1 \log_2 1$$

$$H_{\text{overcast}} = 0$$

Rain:

3 samples, S_4, S_5, S_6 (Yes, No, Yes)

$$P(\text{Yes}) = \frac{2}{3}$$

$$P(\text{No}) = \frac{1}{3}$$

$$\begin{aligned}
 H_{\text{rain}} &= -P_{\text{yes}} \log_2 P_{\text{yes}} + \\
 &\quad - P_{\text{no}} \log_2 P_{\text{no}} \\
 &= -0.67 \log_2 (0.67) + -0.33 \log_2 (0.33) \\
 &= 0.91 \neq 0.92 \\
 &= 0.912
 \end{aligned}$$

⊗ Compute Weighted Entropy

$$\begin{aligned}
 \rightarrow H_{\text{weighted}} &= \frac{\text{No. of Samples}}{\text{Total Samples}} H_{\text{child}} \\
 &= \boxed{\frac{3}{8} * 0} + \boxed{\frac{2}{8} * 0} + \boxed{\frac{3}{8} * 0.912} \\
 &\quad \swarrow \text{sunny} \quad \searrow \text{overcast} \quad \rightarrow \text{rain} \\
 &= 0.344
 \end{aligned}$$

app. to entropy

$$\begin{aligned}
 \rightarrow \text{IG}_{\text{weather}} &= H_{\text{parent}} - H_{\text{child (weighted)}} \\
 &= 1 - 0.344 \\
 \text{Information gain} & \quad \boxed{\text{IG}_{\text{weather}} = 0.656}
 \end{aligned}$$

→ take max IG

⊗ Pick the feature that gives max IG. [argmax]

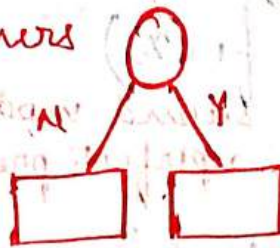
• Difference b/w classifier tree and regression tree is max using procedure strategy

• decision tree are greedy algorithm

Weak learners / model

→ weak model / learners prevents overfitting

• Boosting build strength gradually



$$x_1 \leq 2.5$$

→ Decision stump OR

Decision stub only one level of decision tree

(c) update the residuals

$$r^{(n)} \leftarrow r^{(n)} - \lambda f_b(x^{(n)}, \theta_b)$$

[residuals reduces after each step]

③ final model

$$f(x; \theta) = \sum_{b=1}^B \lambda f_b(x; \theta_b)$$

After each iteration model will get better and better by increasing B . and don't increase λ too much cause otherwise model will overfit

NUMERICAL

y	$f(x)$	$\hat{r} = y - f$
4	0	4
4	0	5
5	0	6
6	0	7

$$f(x) = 0$$

$$f(x) = 5.5$$

x_1	x_2	y
S_1 1.5	2	4
S_2 2	3	5
S_3 3	4	6
S_4 4	5	7

$$f(x) = 5.5$$

Iteration 1

$f(x) = 0$
 $\hat{r}_1 = y$ (initial no prediction)
 (a) fit a model, f_1

Decision stump

$$x_1 \geq 2.5$$

N

Y

$$4.5$$

$$6.5$$

$$S_1 (1, 2) \quad y = 4$$

$$S_2 (2, 3) \quad y = 5$$

$$\frac{4+5}{2} = 4.5$$

$$S_3 (3, 4) \quad y = 6$$

$$S_4 (4, 5) \quad y = 7$$

$$\frac{6+7}{2} = 6.5$$

$$\lambda = 0.1$$

$$f_1(x) \rightarrow x_1$$

x_1	$f_1(x)$
1	4.5
2	4.5
3	6.5
4	6.5

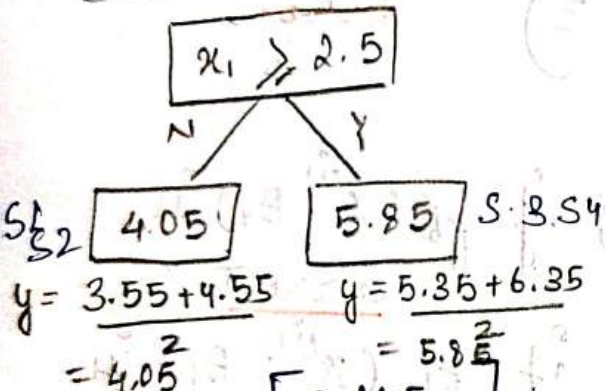
$$r = \begin{bmatrix} 3.55 \\ 4.55 \\ 5.35 \\ 6.35 \end{bmatrix}_{4 \times 1}$$

$$\begin{aligned} \bullet r^{(1)} &= r^{(1)} - \lambda f_1(x^{(1)}) \\ &= 4 - [0.1 \times 4.5] \end{aligned} \quad \left. \begin{array}{l} \bullet r^{(4)} = r^{(4)} - \lambda f_1(x^{(4)}) \\ = 7.0 - [0.1 \times 6.5] \\ = 6.35 \end{array} \right\} \text{UPDATE RESIDUALS}$$

$$\begin{aligned} \bullet r^{(2)} &= r^{(2)} - \lambda f_1(x^{(2)}) \\ &= 5 - [0.1 \times 4.5] \\ &= 4.55 \end{aligned}$$

$$\begin{aligned} r^{(3)} &= r^{(3)} - \lambda f_1(x^{(3)}) \\ &= 6 - [0.1 \times 6.5] \\ &= 5.35 \end{aligned}$$

Iteration 2



x_1	x_2	r
1 1.5	2	3.55
2 2.5	3	4.55
3 3.5	4	5.35
4	5	6.35

$$f_2 \rightarrow r$$

$$f_2(x) \rightarrow x_1$$

x_1	$f_2(x)$
1	4.05
2	4.05
3	5.85
4	5.85

$$r = \begin{bmatrix} 3.145 \\ 4.145 \\ 4.765 \\ 5.765 \end{bmatrix}_{4 \times 1}$$

$$\begin{aligned} 3.145 &= r^{(1)} = 3.55 - [0.1 \times 4.05] \\ 4.145 &= r^{(2)} = 4.55 - [0.1 \times 4.05] \\ 4.765 &= r^{(3)} = 5.35 - [0.1 \times 5.85] \\ 5.765 &= r^{(4)} = 6.35 - [0.1 \times 5.85] \end{aligned}$$

ISLP Boosting

Normal Entropy

$$H = - \sum_{k=1} P_k \log_2 P_k$$

$P_k \rightarrow$ count

$$P_k \rightarrow \frac{\sum_{i \in \text{class } k} W_i}{\sum W_i}$$

05/03

Adaboost Classification

S1

S2

S3

S4

	x_1	x_2	y
	1	2	1
	1.5	1.5	
	2	1	-1
	3	2.5	
	4	4	1
	4.5	3.5	
	5	3	-1

Sample

w (weights)

0.25

0.25

0.25

0.25

① Initialize sample weights
Same steps which wrote

cont.

③ Final model

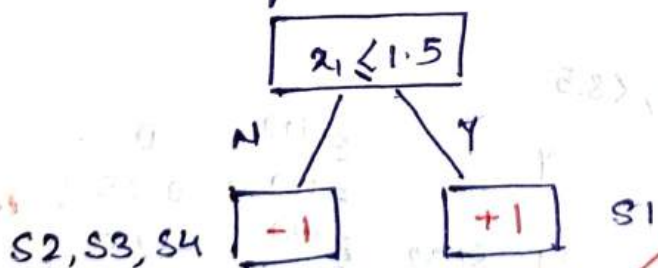
$$F(x; \theta) = \text{sign} \left[\sum_{i=1}^B \lambda_i f_i(x; \theta) \right]$$

Adaboost

$$f_t = \underset{t}{\text{argmin}} \sum_i w^{(i)} I(y^{(i)} \neq f(x^{(i)}; \theta))$$

Possible splits for $x_1 = 1.5, 3, 4.5$

(a)



Total error 0.25

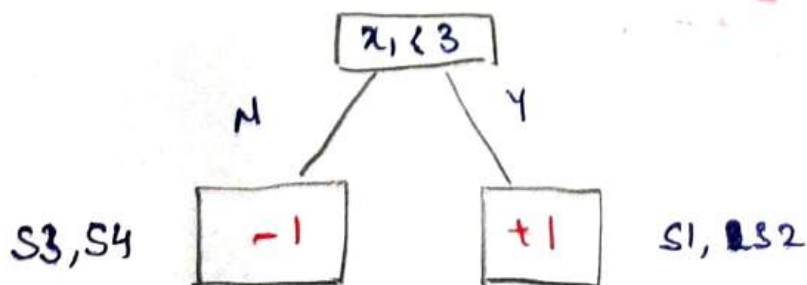
$$(E)^{(1)} = 0$$

$$E^{(2)} = 0$$

at sample 3 the target value not matching so we will give

$$E^{(3)} = 0.25 \text{ [weights]}$$

$$E^{(4)} = 0$$



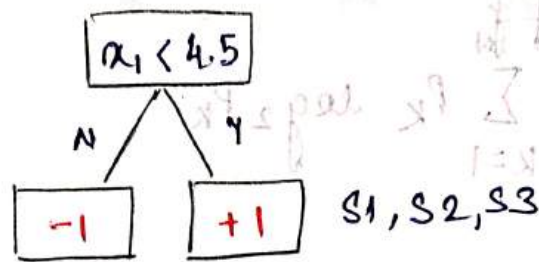
$$E^{(1)} = 0$$

$$E^{(2)} = 0.25$$

$$E^{(3)} = 0.25$$

$$E^{(4)} = 0$$

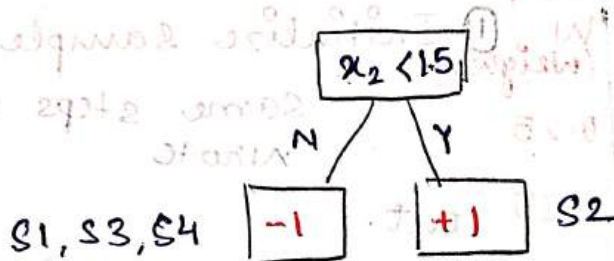
0.50



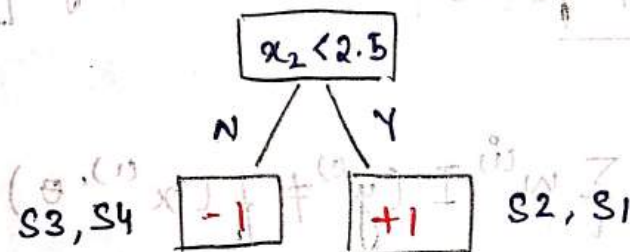
$$\begin{aligned}
 E^{(1)} &= 0 \\
 E^{(2)} &= 0.25 \\
 E^{(3)} &= 0 \\
 E^{(4)} &= 0
 \end{aligned}
 \left. \vphantom{\begin{aligned} E^{(1)} \\ E^{(2)} \\ E^{(3)} \\ E^{(4)} \end{aligned}} \right\} 0.25$$

for x_2

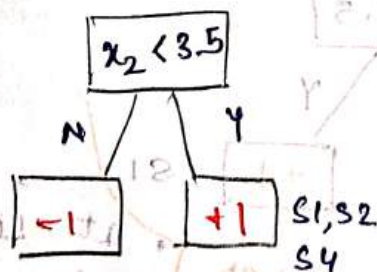
split values = 1.5, 2.5, 3.5



$$\begin{aligned}
 E^{(1)} &= 0.25 \\
 E^{(2)} &= 0.25 \\
 E^{(3)} &= 0.25 \\
 E^{(4)} &= 0
 \end{aligned}
 \left. \vphantom{\begin{aligned} E^{(1)} \\ E^{(2)} \\ E^{(3)} \\ E^{(4)} \end{aligned}} \right\} 0.75$$



$$\begin{aligned}
 E^{(1)} &= 0 \\
 E^{(2)} &= 0.25 \\
 E^{(3)} &= 0.25 \\
 E^{(4)} &= 0
 \end{aligned}
 \left. \vphantom{\begin{aligned} E^{(1)} \\ E^{(2)} \\ E^{(3)} \\ E^{(4)} \end{aligned}} \right\} 0.5$$



$$\begin{aligned}
 E^{(1)} &= 0 \\
 E^{(2)} &= 0.25 \\
 E^{(3)} &= 0.25 \\
 E^{(4)} &= 0.25
 \end{aligned}
 \left. \vphantom{\begin{aligned} E^{(1)} \\ E^{(2)} \\ E^{(3)} \\ E^{(4)} \end{aligned}} \right\} 0.75$$

(b)

$$x_1 < 1.5$$

Model weight

$$\lambda_1 = \frac{1}{2} \ln \left(\frac{1 - e}{e} \right) = \frac{1}{2} \ln \left(\frac{1 - 0.25}{0.25} \right)$$

$$\lambda_1 = 0.5493$$

→ min total error

(c) $w \times e^{-\lambda_1}$

$w \times e^{-\lambda_1}$

$w \times e^{-\lambda_3}$

	old w	new w	Normalized w
S1	0.25	0.144	0.166
S2	0.25	0.144	0.166
S3	0.25	0.4383	0.500
S4	0.25	0.144	0.166
Total		0.865	

new weight
Sum of new weights

$$\frac{0.414}{0.865} = 0.166$$

$$\frac{0.4383}{0.865} = 0.500$$

→ take for next iteration

Iteration - 2

Step 1 Use new weights:

$$W = [0.166, 0.166, 0.500, 0.166]$$

x_1 Possible split of $x_1 = 1.5, 3, 4.5$

$$x_1 \leq 1.5$$



Value and means high Ig (kater)

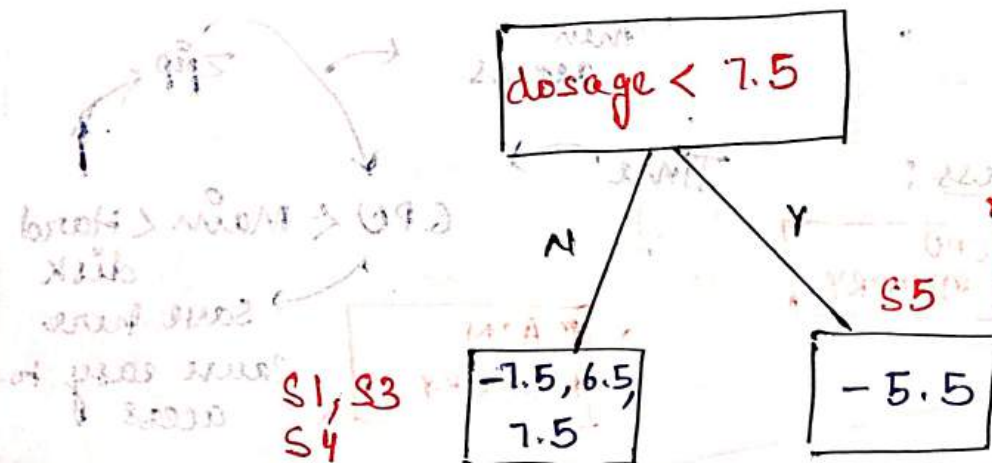
Dosage	Drug Effectiveness	Residuals
S1 - 10	-7	-7.5
S2 - 22.2	-3	-3.5
S3 - 21	7	6.5
S4 - 25	8	7.5
S5 - 5	-5	-5.5
S6 - 20.9	-2	-2.5

Having $\rightarrow S1, S3, S4, S5$

Not $\rightarrow S2, S6$

	dosage	drug Eff	Residuals
S5	5		-5.5
S1	10		-7.5
S3	21		6.5
S4	25		7.5

	dosage	drug	Residual
S2	?	-3	-3.5
S6	?	-2	-2.5



* Assumes missing value on left right when doing IG right & vice versa.

$$IG_{right} = \{-5.5, -3.5, -2.5\}$$

$$IG_{left} = \{-7.5, 6.5, 7.5, -3.5, -2.5\}$$

$$IG_{left} > IG_{right}$$

choose for missing value.

then find default path (the higher path)

- should be group properly that all -ve and +ve go in separate groups.

11/03 # GRADIENT BOOSTING FOR CLASSIFICATION

(prediction) $\hat{y} = 0.66$

eg: matches played
won $\rightarrow 8/2 \rightarrow 4$

	Chest Pain	Good Blood Circulation	Blocked arteries	Heart Disorder	y	(1) log odds	(2) convert to probability
S1	No	No	No	No	0	$0 - 0.66 = -0.66$	
S2	Yes	Yes	Yes	Yes	1	$1 - 0.66 = 0.34$	
S3	Yes	Yes	No	Yes	1	$1 - 0.66 = 0.34$	
S4	Yes	No	No	Yes	1	$1 - 0.66 = 0.34$	
S5	Yes	No	Yes	Yes	1	$1 - 0.66 = 0.34$	
S6	No	Yes	No	No	0	$0 - 0.66 = -0.66$	

Step 1 Initial Prediction

(1) $\log(\text{odds}) = \log\left(\frac{4}{2}\right) \approx 0.7$

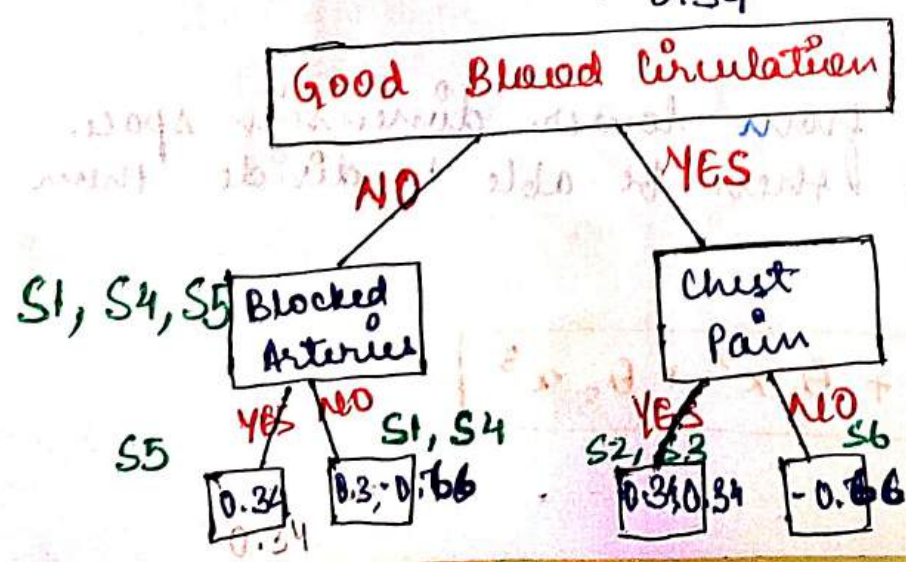
(2) convert this to probability

sigmoid func $\left[P = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}} = \frac{e^{\log(4/2)}}{1 + e^{\log(4/2)}} = 0.66 \right]$

[to make any regression to 0 or 1]
sigmoid func. use

$r = y - \hat{y}$
 $r = [1 - 0.66]$
residuals
0.34

probability value as initial prediction for all samples



S2, S3, S6

convert back into log odds

(3) Predict residuals and convert the leaf values to log (odds)

$$\sum \text{Residual}_i$$

$$\sum_0 [\text{Previous Probability} \times (1 - \text{Previous Probability})]$$

• for 1st leaf = $\frac{0.3}{0.66 \times (1 - 0.66)} \approx 1.43$

• for 2nd leaf = $\frac{0.3 + (-0.7)}{[0.66 \times (1 - 0.66)] + [0.66 \times (1 - 0.66)]}$
 $\approx \frac{-0.4}{0.42} = -0.95$

• for 3rd leaf = $\frac{0.3 + 0.3}{[0.7 \times (1 - 0.7)] + [0.7 \times (1 - 0.7)]} = 1.43$

• for 4th leaf $\rightarrow \frac{-0.7}{[0.7 \times 0.3]} = -3.33$

$$\phi(x) = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1^2 \\ x_1 x_2 \\ x_1 x_3 \\ x_2 x_1 \\ x_2^2 \\ x_1^2 x_2 \end{bmatrix}$$

monomial of x of degree ≤ 3

* in kernel method we can divide the degree till infinite space
* we don't have any restrictions

$$\approx O(d^3) \approx 1000 \ 000 \ 000$$

↳ higher dimension space

• Now have to do gradient descent in 1 Billion dimensional space.

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3$$

$$z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

(converted into higher dim.)

feature map

[A function that converts original data into a new (usually high dim) space]

$$\phi(x) = \begin{bmatrix} x_1 x_1 \\ x_1 x_2 \\ x_1 x_3 \\ x_2 x_1 \\ x_2 x_2 \\ x_2 x_3 \\ x_3 x_1 \\ x_3 x_2 \\ x_3 x_3 \end{bmatrix}_{9 \times 1}$$

$$\phi(z) = \begin{bmatrix} z_1 z_1 \\ z_1 z_2 \\ z_1 z_3 \\ z_2 z_1 \\ z_2 z_2 \\ z_2 z_3 \\ z_3 z_1 \\ z_3 z_2 \\ z_3 z_3 \end{bmatrix}_{9 \times 1}$$

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$z = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\phi(x) = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 4 \\ 6 \\ 3 \\ 6 \\ 9 \end{bmatrix}_{9 \times 1}$$

$$\phi(z) = \begin{bmatrix} 16 \\ 20 \\ 24 \\ 20 \\ 25 \\ 30 \\ 24 \\ 30 \\ 36 \end{bmatrix}_{9 \times 1}$$

kernel
use:
• work in high dim. without actually going there
• Make non linear data become linearly separable.

Now we are doing dot product in higher dim. space

$$\phi(x) \cdot \phi(z) = 16 + 40 + 72 + 40 + 100 + 180 + 72 + 180 + 324 = 1024$$

$$K(x, z) = (\langle x, z \rangle)^2$$

$$K(x, z) = (4 + 10 + 18)^2 = 1024$$

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$z = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$(x_1 z_1 + x_2 z_2 + x_3 z_3)^2$$

→ Polynomial kernel of degree 2

→ Kernel function [By doing in original dim. space we get same value as higher dim. space by using kernel function.]

So, $x \rightarrow \phi(x) \rightarrow$ high dim
↓
dot product

we do!

$$K(x, x') \text{ [no need to compute } \phi(x)]$$

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2} x_1 x_2 \end{bmatrix}_{6 \times 1} \mathbb{R}^6$$

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{(2 \times 1)} \mathbb{R}^2, z = \begin{bmatrix} 3 \\ 4 \end{bmatrix}_{(2 \times 1)} \mathbb{R}^2$$

$$\phi(x) = \begin{bmatrix} 1 \\ 4 \\ 2.828 \\ 1.414 \\ 2.828 \\ 1 \end{bmatrix}_{6 \times 1}$$

$$\phi(z) = \begin{bmatrix} 9 \\ 16 \\ 16.97 \\ 4.242 \\ 5.656 \\ 1 \end{bmatrix}_{6 \times 1}$$

13/03

$$\begin{aligned}
 K(x, z) &= (x + z + 1)^2 \\
 &= (3 + 8 + 1)^2 \\
 &= (12)^2 \\
 &= 144
 \end{aligned}$$

$$\begin{aligned}
 \phi(x) \phi'(z) &= (9) + (16 \times 4) + (16.97 \times 2.828) + (1.414 \times \\
 &\quad 4.242) + (2.828 \times 5.626) + (1 \times 1)
 \end{aligned}$$

$$\begin{aligned}
 &= 9 + 64 + 47.99116 + 5.998188 + 15.910328 \\
 &\neq 1
 \end{aligned}$$

$$= 143.899676 \approx 144$$

17/03

Dataset:
Training:

x_1	x_2	y
1	1	2
3	1	4

Test : $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

By ~~for~~ Linear Regression setup

Hypothesis $h_\theta(x) = \theta^T \phi(x)$ $\theta \in \mathbb{R}^p$
 $\phi(x) \in \mathbb{R}^p$

(1) 1
(2) 3

$$h_\theta(x) = \sum_{i=1}^n \beta_i k(x^{(i)}, x)$$

ϕ is 1 to 2 here in this dataset

- assume β_i learned from training procedure
- we need all β_i 's which are 2 in this dataset as 2 samples.

Let us assume,

$$\beta_1 = 1$$

$$\beta_2 = 1$$

→ another type of kernel

kernel = Radial Basis Function (RBF)

$$K(x, z) = \exp(-\|x - z\|^2)$$

where

$$\|x - z\|^2 = (x_1 - z_1)^2 + (x_2 - z_2)^2$$

solving,

$$\begin{aligned} h_\theta(x) &= \sum_{i=1}^n \beta_i k(x^{(i)}, x) \\ &= \beta_1 k(x^{(1)}, x) + \beta_2 k(x^{(2)}, x) \end{aligned}$$

$$= \beta_1 \exp(-\|x^{(1)} - x\|^2) + \beta_2 \exp(-\|x^{(2)} - x\|^2)$$

Compute,

$$(i) \|x^{(1)} - x\|^2$$

$$\begin{aligned} &= (x_1^{(1)} - x_1)^2 + (x_2^{(1)} - x_2)^2 \\ &= (1 - 2)^2 + (1 - 1)^2 \\ &= (-1)^2 + 0 \\ &= 1 \end{aligned}$$

$$(ii) \|x^{(2)} - x\|^2$$

$$\begin{aligned} &= (x_1^{(2)} - x_1)^2 + (x_2^{(2)} - x_2)^2 \\ &= (3 - 2)^2 + (1 - 1)^2 \\ &= (1)^2 + 0 \\ &= 1 \end{aligned}$$

Substituting β values,

$$\begin{aligned} k(x^{(1)}, x) &= \exp(-\|x^{(1)} - x\|^2) \\ &= \exp(-1) \\ &= 0.367 \end{aligned}$$

$$\begin{aligned} k(x^{(2)}, x) &= \exp(-\|x^{(2)} - x\|^2) \\ &= \exp(-1) \\ &= 0.367 \end{aligned}$$

Final prediction now,

$$\begin{aligned} h_\theta(x) &= \beta_1 k(x^{(1)}, x) + \beta_2 k(x^{(2)}, x) \\ &= 1 \times 0.367 + 1 \times 0.367 \end{aligned}$$

$$h_\theta(x) = 0.734$$

→ these β 's we got from gradient descent in training phase.
→ this is the prediction for new test sample.

Now,

$$p = a \times$$

$$p = a \cdot \frac{a^T b}{a^T a}$$

p = projection matrix (BPM). b

where

$$PM = \frac{a a^T}{a^T a}$$

Projection Matrix in 1D

can't cancel
cause it's a matrix

$$a = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad a^T = [1 \ 2]$$

$$PM = A(A^T A)^{-1} A^T$$

for N -dimensional space

$$\text{let } b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$a = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

horizontal line along the x -axis

find p , i.e. project b onto $a \rightarrow [PM \times b]$

$$PM = \frac{a a^T}{a^T a} = \frac{\begin{bmatrix} 4 \\ 0 \end{bmatrix} \begin{bmatrix} 4 & 0 \end{bmatrix}}{16} = \frac{\begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix}}{16}$$

$$a^T a = \begin{bmatrix} 4 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix} = [16 + 0] = 16$$

$$a a^T = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \begin{bmatrix} 4 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \times \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\frac{aa^T}{a^T a} \Rightarrow \frac{1}{16} \cdot \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix}_{(2 \times 2)} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Dividing $\frac{a^T b}{a^T a}$

PM

$$b \cdot \frac{a^T b}{a^T a}$$

$$\text{PM. } b \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}_{(2 \times 2)} \begin{bmatrix} 2 \\ 3 \end{bmatrix}_{(2 \times 1)}$$

$$\Rightarrow \begin{bmatrix} 2 \times 1 + 0 \times 3 \\ 2 \times 0 + 0 \times 3 \end{bmatrix}_{(2 \times 1)}$$

$$P \Rightarrow \begin{bmatrix} 2 \\ 0 \end{bmatrix}_{(2 \times 1)}$$

Numerical

• Test data

	Actual Label	Predicted Score	Predicted Label
1	① 1	0.85 1	1 1 (OP)
2	0 2	0.6 2	2 ① (PP)
3	① 3	0.7 3	3 1 (OP)
4	1 4	0.4 4	4 0* (PN)
5	0 5	0.55 5	5 ① (PP)
6	① 6	0.5 6	6 1 (OP)
7	0 7	0.65	7 ① (PP)
8	0 8	0.35	8 0 (ON)
9	① 9	0.6	9 1 (OP)
10	0 10	0.2	10 0 (ON)

Threshold
→ 0.5

① Confusion label

② Accuracy

③ Precision

④ Recall/Sensitivity

⑤ F1 Score

⑥ ROC curve

(0.7, 0.5, 0.4, 0.2)

Confusion Matrix

Solⁿ.

	OP	ON
PP	4	3
PN	1	2

TP $\rightarrow$ 4 TN $\rightarrow$ 2

(0.5) Accuracy = $\frac{TP + TN}{TP + TN + FP + FN} = \frac{6}{10} = 0.6$

Precision = $\frac{TP}{TP + FP} = \frac{4}{4 + 3} = \frac{4}{7} = 0.57$

Sensitivity = $\frac{TP}{TP + FN} = \frac{4}{4 + 1} = \frac{4}{5} = 0.8$

Specificity = $\frac{TN}{TN + FP} = \frac{2}{2 + 3} = \frac{2}{5} = \cancel{0.6} 0.4$

F1 Score = $2 \times \frac{0.57 \times 0.8}{0.57 + 0.8} = \frac{0.456}{1.37}$
 $= 2 \times 0.33 = 0.66$

(0.7)

	Actual	Predicted
1.	1	1 (OP)
2.	0	0 (DN)
3.	1	1 (OP)
4.	1	0 (PN)
5.	0	0 (DN)
6.	1	0 (PN)
7.	0	0 (DN)
8.	0	0 (DN)
9.	1	0 (PN)
10.	0	0 (DN)

	OP	ON
PP	2	0
PN	3	5

$$\text{Accuracy} = \frac{OP + ON}{OP + ON + PP + PN} = \frac{2 + 5}{2 + 5 + 3 + 0} = \frac{7}{10} = 0.7$$

$$\text{Precision} = \frac{OP}{OP + PP} = \frac{2}{2 + 0} = 1$$

$$\text{Sensitivity} = \frac{OP}{OP + PN} = \frac{2}{2 + 3} = \frac{2}{5} = 0.4$$

$$\text{Specificity} = \frac{ON}{ON + PP} = \frac{0}{0 + 5} = 0$$

$$F1 = 2 \times \frac{\text{precision} \times \text{sensitivity}}{\text{precision} + \text{sensitivity}} = 2 \times \frac{1 \times 0.4}{1 + 0.4} = 2 \times \frac{0.4}{1.4} = \frac{0.8}{1.4} = 0.57 \approx 0.6$$

(0.4)

	Actual	Predicted	
1.	1	1	(TP)
2.	0	1	(FP)
3.	1	1	(TP)
4.	1	1	(TP)
5.	0	1	(FP)
6.	1	1	(TP)
7.	0	1	(FP)
8.	0	0	(TN)
9.	1	1	(TP)
10	0	0	(TN)

TP	FP
5	3
2	0
TN	FN

$$\text{Accuracy} \rightarrow \frac{TP + TN}{\text{Total}} = \frac{7}{10} = 0.7$$

$$\text{Sensitivity} \rightarrow \frac{TP}{TP + FN} = \frac{5}{5 + 0} = 1$$

$$\text{Specificity} \rightarrow \frac{TN}{TN + FP} = \frac{2}{2 + 3} = \frac{2}{5} = 0.4$$

$$\text{Precision} \rightarrow \frac{TP}{TP + FP} = \frac{5}{5 + 3} = \frac{5}{8} = 0.625$$

$$F1 \text{ Score} \rightarrow 2 \times \frac{0.625 \times 1}{0.625 + 1} = 2 \times \frac{0.625}{1.625}$$

$$\Rightarrow \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 0.77$$

0.2

	Actual	Predicted	
1.0 x 1 = 1	1	1	(TP)
1.0 + 11.			
2. x 6 = 0	0	1	(FP)
3.	1	1	(TP)
4.	1	1	(TP)
5.	0	1	(FP)
6.	1	1	(TP)
7.	0	1	(FP)
8.	0	1	(FP)
9.	1	1	(TP)
10	0	0	(TN)

TP	FP
5	5
00	00
FN	TN

$$\text{Accuracy} \rightarrow \frac{TP + FP}{\text{Total}} = \frac{10}{10} = 1$$

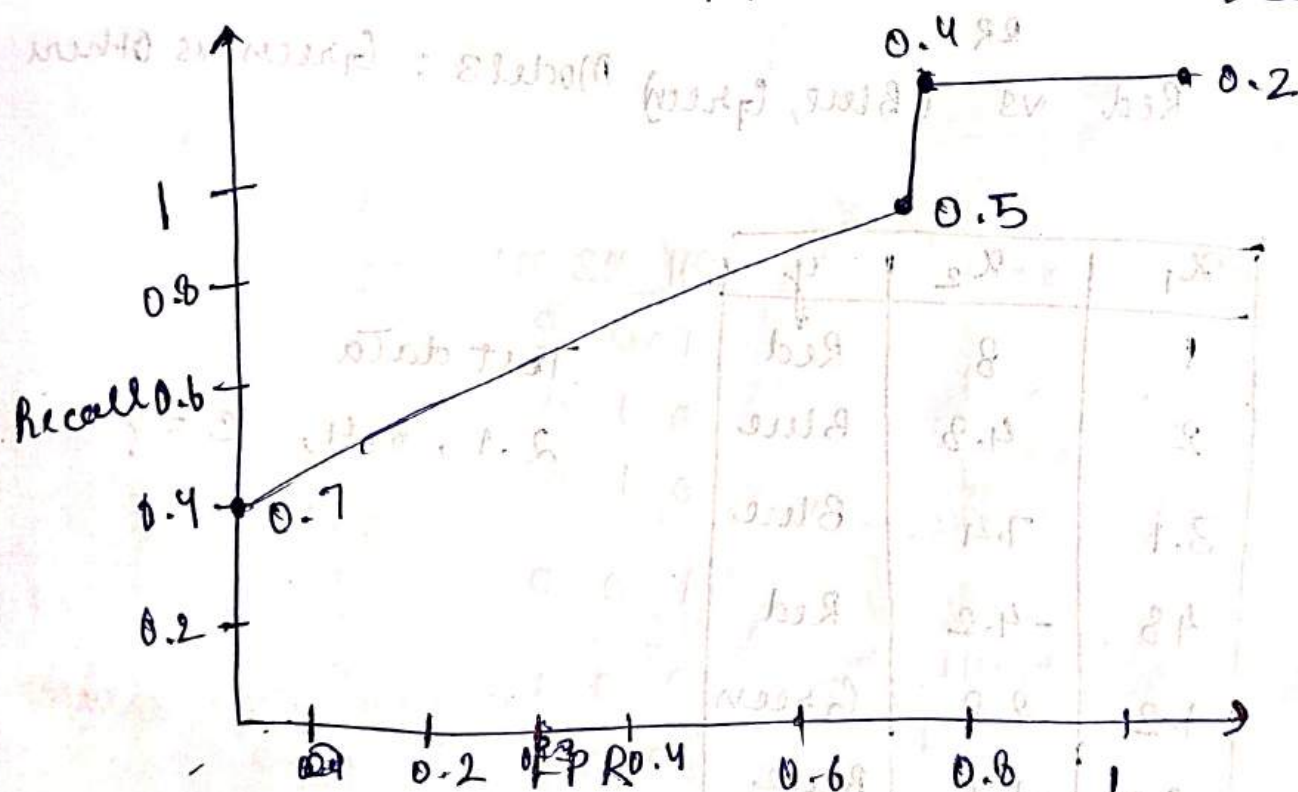
$$\text{Sensitivity} \rightarrow \frac{TP}{TP + FN} = \frac{5}{5 + 0} = 1$$

$$\text{Specificity} \rightarrow \frac{TN}{TN + FP} = \frac{0}{0 + 5} = 0$$

$$\text{Precision} \rightarrow \frac{TP}{TP + FP} = \frac{5}{5 + 5} = 0.5$$

$$F1 \text{ score} \rightarrow \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.55 \times 1}{0.55 + 1}$$

$$= 2 \times \frac{1 \times 1}{1 + 1} = 2 \times \frac{1}{2} = 1$$



0.5 (0.6, 0.8)

0.7 (

FPR $\rightarrow 1 - \text{Specificity}$

=

Specificity

FPR

Threshold

0.2

0.4

0.5

0.7

0.2

0.4

0.8

0.4

0.2

0.4

0.4

1

0.8

0.6

0.6

0

① Correlation and Covariance

X	Y
1	2
2	4
3	5
4	4
5	5

X: Hours studied

Y: Exam Score

(i) Correlation $\rightarrow$

$$r = \frac{\text{COV}(X, Y)}{\sigma_X \sigma_Y}$$

(ii) $\text{COV}(X, Y)$

$$= \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

(iii) $\sigma_X = \sqrt{\frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})^2}$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x^{(i)}$$

$\sigma_X = \frac{\sqrt{2}}{1.414} = 1.09$

$\bar{x} = 3$

$\bar{y} = 4$

• $\text{COV} = \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$

$$= \frac{1}{5} [(1-3)(2-4) + (2-3)(4-4) + (3-3)(5-4) + (4-3)(4-4) + (5-3)(5-4)]$$

$$= \frac{1}{5} [(-2)(-2) + (-1)(0) + (0)(1) + (1)(0) + (2)(1)]$$

$$= \frac{1}{5} [4 + 0 + 0 + 0 + 2]$$

$$= \frac{6}{5} \Rightarrow 1.2$$

• Correlation $\rightarrow \frac{1.2}{1.41 \times 1.09} = \frac{1.2}{1.53} \Rightarrow 0.78$

Eigen Values and Eigen Vectors

$$\boxed{\det(A - \lambda I) = 0}$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix}$$

• Determinant $\Rightarrow$

$$(2 - \lambda)(2 - \lambda) - 1(1) = 0$$

$$(2 - \lambda)^2 - 1 = 0$$

$$4 - 4\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

$$\boxed{\lambda = 3, 1}$$

$$\textcircled{1} \lambda_1 = 3$$

$$(A - 3I)v = 0$$

$$3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow (-1)x + (1)y = 0$$

$$(1)x + (-1)y = 0$$

$$\Rightarrow -x + y = 0$$

$$\Rightarrow \boxed{x = y}$$

$$\bullet \text{ if } x = 1 \text{ then } y = 1$$

$$\textcircled{2} \lambda_2 = 1$$

$$(A - 1I)v = 0$$

$$1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$(A - 3I) = \begin{bmatrix} 2-1 & 1-0 \\ 1-0 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$(A - I) \mathbf{v} \Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow x = -y$$

• if $x = 1$ then $y = -1$

PCA

$$X = \begin{bmatrix} x_1 & x_2 & x_3 \\ 2 & 0 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix} \quad d = 3$$

Step 1 Mean centering

$$\mu_{x_1} = 1.5$$

$$\mu_{x_2} = 1.5$$

$$\mu_{x_3} = 1.5$$

$$\mu = [1.5 \quad 1.5 \quad 1.5]$$

$$X_{\text{centered}} = \begin{bmatrix} 0.5 & -1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 \\ 1.5 & -0.5 & 0.5 \\ -0.5 & 1.5 & 0.5 \end{bmatrix}$$

$$X_c = X - \mu$$

~~Subtract~~
by respective
mean values

Step 2 Covariance Matrix

$$\begin{bmatrix} \text{COV}(x_1, x_1) & \text{COV}(x_1, x_2) & \text{COV}(x_1, x_3) \\ \text{COV}(x_2, x_1) & \text{COV}(x_2, x_2) & \text{COV}(x_2, x_3) \\ \text{COV}(x_3, x_1) & \text{COV}(x_3, x_2) & \text{COV}(x_3, x_3) \end{bmatrix}$$

$$C = \frac{1}{n} X_C^T X_C$$

$$(3 \times 4) \quad (4 \times 3)$$

$$C = \begin{bmatrix} 1.25 & -0.75 & 0.5 \\ -0.75 & 1.25 & 0.5 \\ 0.5 & 0.5 & 0.25 \end{bmatrix} \quad (3 \times 3)$$

Step 3 Eigen values and Eigen vectors

$$\lambda_1 = 2.0, \quad \lambda_2 = 0.75, \quad \lambda_3 = 0$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Step 4 Dimensionality Reduction

$$3D \rightarrow 2D$$

- Take top 2 eigen vectors

$$W_{2D} = \begin{bmatrix} u_1 & u_2 \end{bmatrix}$$

- Project the data points

$$Z = X_C W_{2D}$$

$$= \begin{bmatrix} \end{bmatrix}$$

Each point will be of 2D

$$3D \rightarrow 1D$$

Take top 1 eigen vector
 $W_{1D} = [v_1]$

• Total variance
 $= 2 + 0.75 + 0$
 $= 2.75$

$$PC1 = \frac{2}{2.75} \approx 73\%$$

$$PC2 = \frac{0.75}{2.75} \approx 27\%$$

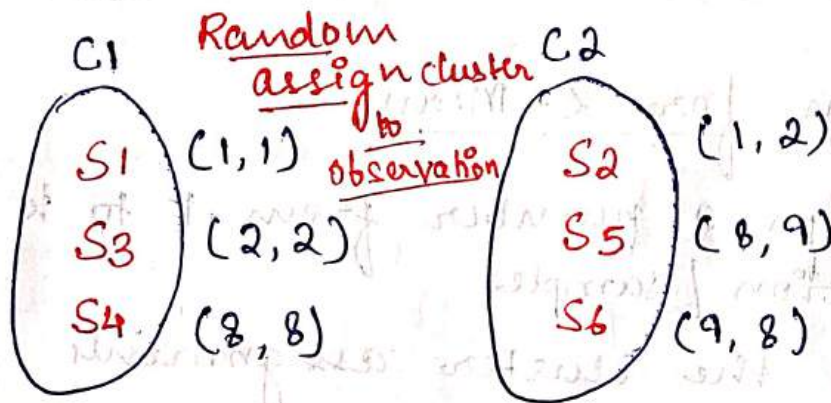
$$PC3 = \frac{0}{2.75} = 0\%$$

NUMERICAL

	x_1	x_2
S_1	1	1
S_2	1	2
S_3	2	2
S_4	8	8
S_5	8	9
S_6	9	8

$k=2$

Step 1



Step 2

avg. vector of feature mean
Compute the centroid for all the clusters (C_1, C_2)

$$C_1 \Rightarrow \left[\left(\frac{1+2+8}{3} \right), \left(\frac{1+2+8}{3} \right) \right] = \left(\frac{11}{3}, \frac{11}{3} \right) = (3.66, 3.66)$$

$$C_2 \Rightarrow \left[\left(\frac{1+8+9}{3} \right), \left(\frac{2+9+8}{3} \right) \right] = \left(6, \frac{19}{3} \right) = (6, 6.33)$$

Step 3

Process each point and assign the cluster based on the minimum distance to the centroid.

- $S_1 \rightarrow$ distance to centroid (C_1)
 $\Rightarrow (1, 1) \Rightarrow$

$$\begin{aligned}
 r_1 &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(3.67 - 1)^2 + (3.67 - 1)^2} \\
 &= \sqrt{(2.67)^2 + (2.67)^2} \\
 &= \cancel{7.12} \ 3.78 \quad (\text{cluster 1 min})
 \end{aligned}$$

- $S_1 \rightarrow$ distance to centroid (C_2)
 $\Rightarrow (1, 1) \Rightarrow$

$$\begin{aligned}
 r_2 &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(6 - 1)^2 + (6.34 - 1)^2} \\
 &= \sqrt{(5)^2 + (5.34)^2} \\
 &= \sqrt{25 + 28.5156} \\
 &= 7.31
 \end{aligned}$$

• ~~$S_2 \rightarrow \sqrt{(3.67 - 1)^2}$~~

• $S_2 \rightarrow \sqrt{(3.67 - 2)^2 + (3.67 - 2)^2} \quad (C_1)$
 $= \sqrt{2.78^2}$
 $= 2.86 \quad (\text{cluster 1 min})$

• $S_3 \rightarrow \sqrt{(6 - 2)^2 + (6.34 - 2)^2} \quad (C_2)$
 $= \sqrt{(4)^2 + (4.34)^2}$
 $= 5.9$

$$S_4 \rightarrow (8, 8) \quad (C_1)$$

$$= \sqrt{(3.67-8)^2 + (3.67-8)^2}$$

$$= \sqrt{(5.67)^2 + (5.67)^2}$$

$$= 8.02$$

$$S_4 \rightarrow (8, 8) \quad (C_2)$$

$$= \sqrt{(6-8)^2 + (6.34-8)^2}$$

$$= \sqrt{(2)^2 + (2.34)^2}$$

$$= 3.079$$

(cluster 2)

$$S_2 \rightarrow (1, 2) \quad (C_1)$$

$$= \sqrt{(3.67-1)^2 + (3.67-2)^2}$$

$$= \sqrt{(2.67)^2 + (1.67)^2}$$

$$= \sqrt{7.13 + 2.79}$$

$$= 3.14$$

(cluster 1)

$$S_2 \rightarrow (1, 2) \quad (C_2)$$

$$= \sqrt{(6-1)^2 + (6.34-2)^2}$$

$$= \sqrt{25 + 18.84}$$

$$= 6.62$$

$$S_5 \rightarrow (8, 9) \quad (C_1)$$

$$= \sqrt{(3.67-8)^2 + (3.67-9)^2}$$

$$= \sqrt{(5.67)^2 + (5.67)^2} = 8.02$$

$$S5 \rightarrow (8, 9) \quad C2$$

$$= \sqrt{(6-8)^2 + (6.34-9)^2}$$

$$= \sqrt{(2)^2 + (2.34)^2}$$

$$= \sqrt{4 + 11.15}$$

$$= \sqrt{15.15} = 3.9$$

(Cluster C2)

~~$$S5 \rightarrow (8, 9)$$~~

= 0

$$S6 \rightarrow (9, 8) \quad C1$$

$$= \sqrt{(3.67-9)^2 + (3.67-8)^2}$$

$$= 8.26$$

$$S6 \rightarrow (9, 8) \quad C2$$

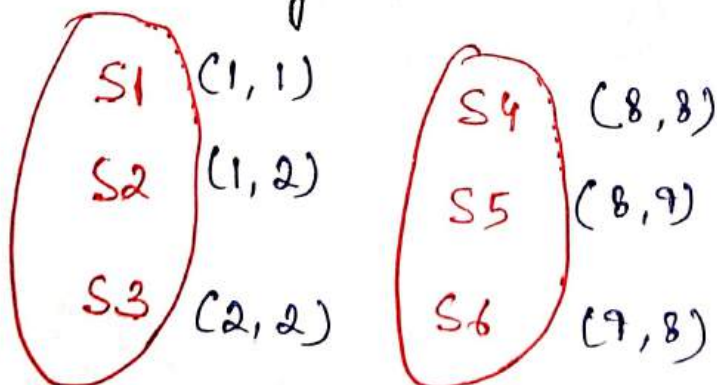
$$= \sqrt{(6-9)^2 + (6.34-8)^2}$$

$$= \sqrt{9 + (2.34)^2}$$

$$= 3.8$$

(Cluster C2)

* New assignment :



Compute same steps again :

Ques)

4 SAMPLES

A

2

B

5

C

8

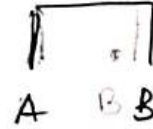
D

	A	B	C	D
A	0	1	4	7
B	1	0	3	6
C	4	3	0	3
D	7	6	3	0

$$\{(A, B), (A, C), (A, D), (B, C), (B, D), (C, D)\}$$

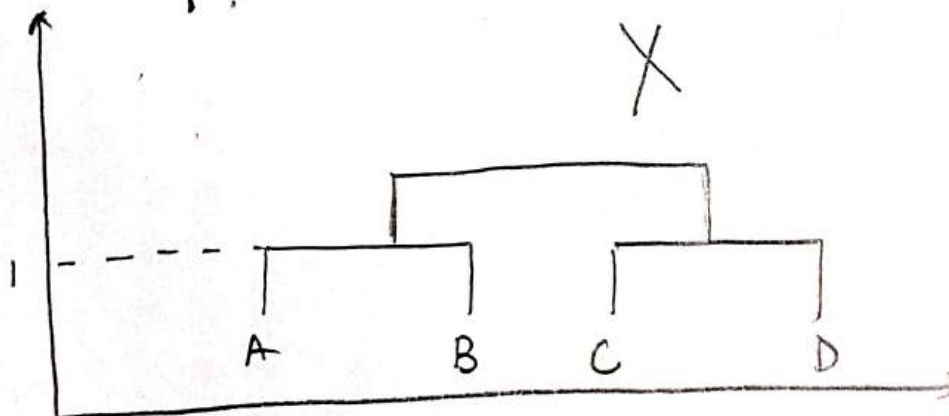
- The distance b/w A and B is 1 which is least
- So make A and B cluster $\{A, B\} \{C\} \{D\}$

	$\{A, B\}$	$\{C\}$	$\{D\}$
$\{A, B\}$	0	4	7
$\{C\}$	4	0	3
$\{D\}$	7	3	0



- The distance b/w C and D is 3 which is least
- So make C and D cluster $\{A, B\} \{C, D\}$

	$\{A, B\}$	$\{C, D\}$
$\{A, B\}$	0	7
$\{C, D\}$	7	0



Y axis

DENDROGRAM complete linkage

dissimilarity score

